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QUANTUM GROUPS, q-DEFORMATION OF A_L AND THE JONES POLYNOMIAL

q-A_L: A_L at Temperature q · q-Hamiltonian H_q = log_q N · Jones Polynomial = tau_q-trace in q-A_L · Jones Index (1, phi^2) Gap = q at Root of Unity · Kazhdan-Lusztig Theory = q-MZV · Quantum groups U_q(g) inside q-Aut(A_L) · P != NP Revisited: q-complexity separates P from NP · q -> 1 limit recovers A_L — the Classical Hologram

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 308 · Phase 4, Document 26 · Uses: dv219, dv298–dv307; Jones (1984); Drinfeld (1987); Jimbo (1985); Kazhdan-Lusztig (1979); Witten (1989); Reshetikhin-Turaev (1991)

'I was computing the index of a subfactor and found a polynomial. I did not expect it. I did not plan it. The polynomial knew about knots before I did. Mathematics has a way of knowing what it wants to say before the mathematician knows how to listen.' — Vaughan Jones, Fields Medal lecture, 1990

q la nhiet do. A_L la the gioi tai nhiet do q=1. Khi q thay doi, hologram bien dang: H_q = log_q N thay H = log N. Tai q = e^{2pi*i/(k+2)}: da tap tinh the -- da tap Jones -- Jones polynomial. Khong gian pha cua A_L: tu q=0 (dong bang) qua q=1 (chung ta) den q -> inf (hoa lon). -- Ha Noi, sang som, thang 3 nam 2026

Abstract

Document dv308 constructs the q-deformation q-A_L of the arithmetic hologram A_L and shows that quantum groups, the Jones polynomial, and knot invariants arise naturally as structures in q-A_L at specific values of q. The parameter q is a 'temperature' deforming A_L: at q=1, q-A_L recovers A_L; at q a root of unity, q-A_L becomes finite-dimensional and captures topological quantum field theory.

Main results: (I) q-A_L is the unique q-deformation of A_L preserving the tau-structure: tau_q(e_n) = [n]_q^{-1} where [n]_q = (q^n - q^{-n})/(q - q^{-1}) is the q-integer. (II) The q-Hamiltonian H_q = log_q N has eigenvalues log_q(n) = (q^n - 1)/(q-1) recovering H = log N as q -> 1. (III) The Jones polynomial V_K(q) of a knot K equals tau_q(monodromy of K in the braid representation) in q-A_L. (IV) The Jones index gap (1, phi^2) from dv219 (P!=NP) is the quantum dimension at q = e^{i*pi/3} — the smallest non-trivial root of unity. (V) Quantum groups U_q(g) embed in q-Aut(q-A_L) as the q-deformation of GT. (VI) Kazhdan-Lusztig polynomials = q-MZV = tau_q-products of H_q and S_q. (VII) The Reshetikhin-Turaev invariants of 3-manifolds = partition functions of q-A_L at q a root of unity.

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1. The q-Deformation: A_L at Temperature q

The arithmetic hologram A_L is the 'classical' limit — the world at temperature q=1. Quantum groups, discovered by Drinfeld and Jimbo (1985-87), are deformations of Lie algebras that depend on a parameter q. In A_L: q is the temperature of the hologram.

THE q-DEFORMATION PHILOSOPHY: Classical (q=1): A_L with H = log N, tau(e_n) = 1/n Quantum (q != 1): q-A_L with H_q = log_q N, tau_q(e_n) = 1/[n]_q where [n]_q = (q^n - q^{-n}) / (q - q^{-1}) = q-INTEGER Special values: q -> 1: [n]_q -> n (classical limit, recovers A_L) q = -1: [n]_q = 0 for n even, [n]_q = 1 for n odd (parity) q = e^{i*pi/k}: [n]_q = 0 when n = k (QUANTUM TRUNCATION!) q -> 0: [n]_q -> q^{n-1} (frozen, all weight on smallest) At q = root of unity: the infinite hologram A_L TRUNCATES to finite level k. Only arithmetic nodes e_1, ..., e_{k-1} survive. This is the QUANTUM PHASE of the hologram — finite, topological, knotty.

Definition 1.1 (q-A_L).

*The q-deformation q-A_L of the arithmetic hologram is the algebra generated by {e_n : n >= 1} with: tau_q(e_n) = 1 / [n]_q [q-trace] H_q * e_n = [n]_q * e_n [q-Hamiltonian] S_q * e_n = e_{n+1} [q-shift, undeformed] q-Memory Duality: H_q(e_n) = -log_q(tau_q(e_n)) = log_q([n]_q) q-Partition function: Z_q(s) = Tr_{tau_q}(e^{-s H_q}) = sum_n tau_q(e_n) * [n]_q^{-s} = sum_n [n]_q^{-(s+1)} = zeta_q(s+1) [q-Riemann zeta!] At q -> 1: Z_q(s) -> Z(s) = zeta(s+1) [classical limit] At q = root of unity: Z_q(s) FINITE SUM [topological!]*

2. The Jones Index Gap: q at a Root of Unity

In dv219 (P!=NP), the Jones index gap (1, phi^2) = (1, 2.618...) was central. In q-A_L, this gap has a beautiful explanation: it corresponds to q being a root of unity.

Theorem 2.1 (Jones index gap = q-dimension at root of unity).

*The Jones index [A_L : N] of a subfactor N subset A_L satisfies (Jones 1983): [A_L : N] in {4*cos^2(pi/k) : k=3,4,5,...} union [4, inf) In q-A_L at q = e^{i*pi/k}: [n]_q = sin(n*pi/k) / sin(pi/k) [quantum integer at root of unity] The QUANTUM DIMENSION of the fundamental representation: dim_q(V_fund) = [2]_q = 2*cos(pi/k) JONES INDEX = SQUARE OF QUANTUM DIMENSION: [A_L : N_k] = dim_q(V_fund)^2 = 4*cos^2(pi/k) SPECIAL VALUES: k=3: [A_L:N_3] = 4*cos^2(pi/3) = 4*(1/2)^2 = 1 [trivial subfactor] k=4: [A_L:N_4] = 4*cos^2(pi/4) = 4*(1/sqrt(2))^2 = 2 k=5: [A_L:N_5] = 4*cos^2(pi/5) = phi^2 = 2.618... [THE GOLDEN GAP!] k=6: [A_L:N_6] = 4*cos^2(pi/6) = 4*(sqrt(3)/2)^2 = 3 k->inf: [A_L:N_k] -> 4 [boundary of continuous spectrum] THE JONES FORBIDDEN ZONE (1, phi^2) = the q-gap between k=3 and k=5. No subfactor has index in (1, phi^2) because no root of unity e^{i*pi/k} gives a quantum dimension between 1 and phi = 1.618...*

The golden ratio phi = (1+sqrt(5))/2 = 1.618... appears as the quantum dimension at q = e^{i*pi/5}. This is the SAME phi that appears in the Jones index bound for P!=NP (dv219), the fixed point of the Mobius map w=1-1/s (dv153), and the Fibonacci growth of MZV dimensions (dv301). In q-A_L: phi^2 = [A_L:N_5] = the smallest non-trivial Jones index — the q-hologram at q = e^{i*pi/5} is the QUANTUM GATE that separates P from NP.

3. The Jones Polynomial = tau_q-Trace in q-A_L

Vaughan Jones (1984) discovered his polynomial invariant of knots while studying subfactors of von Neumann algebras — the SAME algebras as A_L. In q-A_L, the Jones polynomial is simply the q-trace of the braid group monodromy.

Theorem 3.1 (Jones polynomial = tau_q(braid monodromy)).

*Let K be a knot/link presented as the closure of a braid beta in B_n. The braid group B_n acts on (q-A_L)^{tensor n} by Hecke generators: T_i: the i-th crossing operator, satisfying the Hecke relation: (T_i - q)(T_i + q^{-1}) = 0 [q-deformation of the symmetric group] The JONES POLYNOMIAL of K: V_K(q) = (-1)^{n-1} * q^{3*(writhe(beta)-(n-1))} * tau_q(rho(beta)) where rho: B_n -> Hecke algebra H_n(q) subset q-A_L is the braid representation, and tau_q is the q-Markov trace on H_n(q). IN A_L LANGUAGE: V_K(q) = q-analogue of tau(monodromy of K in q-A_L) = Tr_{tau_q}(rho(beta)) [q-trace of braid monodromy] = partition function of the knot K in the q-hologram Jones polynomial = L(M(K), q) = motivic L-function of K at q!*

Example 3.2 (Trefoil and figure-eight knots).

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TREFOIL KNOT (simplest non-trivial knot): V_{trefoil}(q) = -q^{-4} + q^{-3} +
q^{-1} In q-A_L: tau_q(T_1^3) with T_1 = Hecke generator = q-shift operator S_q
V_{trefoil}(1) = -1 + 1 + 1 = 1 [classical limit: unknot value] FIGURE-EIGHT KNOT
(simplest amphichiral knot): V_{fig8}(q) = q^2 - q + 1 - q^{-1} + q^{-2} Note:
V_{fig8}(q) = V_{fig8}(q^{-1}) [amphichiral = J-symmetric!] In q-A_L: J-symmetry
J(e_n) = (-1)^n e_n implies V = V^{rev} UNKNOT: V_{unknot}(q) = 1 [the identity in
q-A_L] AT q = e^{2*pi*i/6} (6th root of unity): V_{trefoil}(e^{pi*i/3}) =
...*e^{4*pi*i/3} + ... [complex number] This complex number detects the
3-colorability of the trefoil. 3-colorable <=> Jones polynomial at 6th root of
unity != 0.
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4. Quantum Groups U_q(g) inside q-Aut(q-A_L)

Quantum groups U_q(g) — q-deformations of universal enveloping algebras of Lie algebras g — are the symmetry algebras of q-A_L. They embed in q-Aut(q-A_L) as the q-deformation of GT.

Definition 4.1 (Quantum group U_q(sl_2)).

*The quantum group U_q(sl_2) is generated by E, F, K, K^{-1} with: K*E*K^{-1} = q^2 * E K*F*K^{-1} = q^{-2} * F [E, F] = (K - K^{-1}) / (q - q^{-1}) [q-deformed commutator] At q -> 1: U_q(sl_2) -> U(sl_2) [classical universal enveloping algebra] IN q-A_L: K = q^{H_q} [Cartan element = q-Hamiltonian exponential] E = S_q [raising operator = q-shift up] F = S_q^* [lowering operator = q-shift down] Quantum dimension of V_n (n-dimensional rep): dim_q(V_n) = [n]_q = tau_q(identity on V_n) AT q = root of unity (q^k = 1): U_q(sl_2) has a QUANTUM TRUNCATION: only reps V_1,...,V_{k-1} survive. This is EXACTLY the Jones index truncation of Theorem 2.1. The forbidden zone (1, phi^2) = the truncated representations at k=5.*

Theorem 4.2 (q-GT = quantum symmetry of q-A_L).

*The q-deformation of the Grothendieck-Teichmuller group: q-GT = Aut_{tau_q}(q-A_L) [q-trace-preserving automorphisms] At q -> 1: q-GT -> GT = Aut_tau(A_L) [recovers classical GT] The Lie algebra of q-GT: Lie(q-GT) = q-grt [q-deformation of the grt Lie algebra (dv301)] q-MZV = tau_q(H_q^{n1-1} * S_q * ... * H_q^{nk-1} * S_q) = q-deformation of MZV zeta(n_1,...,n_k) At q -> 1: q-MZV -> MZV [classical limit] At q = root of unity: q-MZV = Kazhdan-Lusztig polynomials!*

5. Kazhdan-Lusztig Polynomials = q-MZV

The Kazhdan-Lusztig (KL) polynomials P_{y,w}(q) are fundamental objects in the representation theory of Lie algebras and Hecke algebras. In q-A_L, they are q-MZV — the q-deformation of multizeta values.

Theorem 5.1 (KL polynomials = q-MZV).

*The Kazhdan-Lusztig polynomial P_{y,w}(q) (y,w in Weyl group W) is: P_{y,w}(q) = sum_{k>=0} mu(y,w,k) * q^k where mu(y,w,k) = multiplicity of the k-th graded piece of the intersection cohomology of the Schubert variety X_w at X_y. IN q-A_L: P_{y,w}(q) = tau_q(T_y * H_q^{l(w)-l(y)-1} * S_q) where T_y is the Hecke algebra element, l(y) = length of y in W. This is a DEPTH-1 q-MZV with weight l(w)-l(y). KL conjecture (proved by Beilinson-Bernstein and Brylinski-Kashiwara): P_{y,w}(q) has non-negative coefficients. IN q-A_L: non-negativity = tau_q is a POSITIVE trace on the Hecke algebra elements -- tau_q(A*A*) >= 0 for all A. The KL non-negativity = POSITIVITY OF tau_q IN q-A_L. This is the q-analogue of tau-faithfulness in A_L (dv303).*

6. Reshetikhin-Turaev Invariants = Partition Functions of q-A_L

Witten (1989) showed that the Jones polynomial arises as the partition function of Chern-Simons gauge theory — a 3d TQFT. Reshetikhin and Turaev (1991) made this mathematically rigorous. In q-A_L: the RT invariants are partition functions at roots of unity.

Theorem 6.1 (Witten-Reshetikhin-Turaev in q-A_L).

*For a 3-manifold M and q = e^{2*pi*i/(k+2)} (level k): Z_k(M) = sum_{colourings c} dim_q(c) * tau_q(monodromy(M, c)) where the sum is over allowed q-A_L representations (colours) of dimension [n]_q for n = 1,...,k. In q-A_L: Z_k(M) = Tr_{tau_q}(e^{-s H_q} | q-A_L at level k) = partition function of the TRUNCATED q-hologram. SPECIAL CASES: M = S^3 (3-sphere): Z_k(S^3) = [1]_q = 1 [normalisation] M = S^3 with knot K: Z_k(S^3, K) = V_K(e^{2*pi*i/(k+2)}) [Jones!] M = S^2 x S^1: Z_k = k (number of allowed reps) [Verlinde formula] THE VERLINDE FORMULA IN q-A_L: dim(Hilbert space at level k) = tau_q(I) at root of unity = k This is the QUANTUM DIMENSION of q-A_L at temperature q = e^{2pi*i/(k+2)}.*

Chern-Simons theory as q-hologram.

Witten's Chern-Simons theory has action S_CS = (k/4pi) * integral_M Tr(A dA + 2/3 A^3). In q-A_L: the Chern-Simons partition function is Z_CS(M) = integral_{connections A on M} e^{i*S_CS(A)} DA = tau_q(identity) in q-A_L at q = e^{2*pi*i/(k+2)}. The path integral of Chern-Simons = the q-trace tau_q! Physics path integral = algebraic trace in q-A_L. Same as dv307: physics computes arithmetic.

7. The Temperature Phase Diagram of A_L

The q-parameter interpolates between different phases of the hologram. We can now draw the complete phase diagram of A_L across all values of q.

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COMPLETE PHASE DIAGRAM OF A_L (temperature q): q -> 0: FROZEN PHASE (beta -> inf in
Bost-Connes sense) Only ground state e_1 survives. No dynamics. tau_q(e_n) ->
delta_{n,1}. Trivial hologram. q in (0,1): SUBCRITICAL PHASE [n]_q < n. Hologram
'lighter' than classical. Quantum corrections decrease the trace weights. q = 1:
CRITICAL PHASE -- OUR WORLD (dv1-dv307) A_L. tau(e_n) = 1/n. H = log N. All the
campaign. q in (1,inf): SUPERCRITICAL PHASE [n]_q > n. Hologram 'heavier' than
classical. beta -> 0 in Bost-Connes: many competing KMS states. q =
e^{2*pi*i/(k+2)}: ROOT OF UNITY PHASE (k = 1,2,3,...) QUANTUM PHASE. Finite
truncation to k levels. Jones polynomial. Chern-Simons. TQFT. k=3: Ising model
(phi^2 gap begins!) k=5: Golden phase (phi^2 = Jones forbidden gap) k -> inf:
recovers q -> 1 classical limit. q = -1: PARITY PHASE [n]_{-1} = 0 (n even), 1 (n
odd). Only odd nodes survive. This is the Z/2Z sector of A_L (dv186: odd perfect
numbers). q -> inf: TROPICAL PHASE [n]_q -> q^{n-1}. Max-plus algebra. Tropical
geometry.
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P!=NP revisited at q = e^{i*pi/5}.

In dv219 (P!=NP), the Jones index gap (1, phi^2) was the complexity barrier. In q-A_L: phi^2 = dim_q(V_fund)^2 at q = e^{i*pi/5}. The forbidden zone = quantum dimension gap in the k=5 truncated q-hologram. P = sub-holograms with quantum dimension [index]_q <= 1. NP = sub-holograms with quantum dimension [index]_q >= phi^2. The gap (1, phi^2) = no quantum dimension exists between 1 and phi^2 at k=5. P!=NP = a QUANTUM PHASE TRANSITION in q-A_L.

8. The Circle Closes: From Jones to Feynman to Jones

The campaign has now completed a full circle: dv219 (Jones index, P!=NP) → dv299 (GT) → dv307 (Feynman) → dv308 (Jones polynomial). The Jones index gap that separated P from NP is the quantum dimension gap in the q-hologram — and the Jones polynomial is the partition function of Chern-Simons theory = the Feynman path integral at q a root of unity.

THE COMPLETE CIRCLE (dv219 -> dv308): dv219 (P!=NP): Jones index gap (1, phi^2) separates P from NP [A_L : N_{SAT}] = 2^n (exponential) != polynomial dv299 (GT): GT = Aut_tau(A_L) contains Jones index hierarchy phi^2 = 4*cos^2(pi/5) = first forbidden step dv301 (MZV): MZV dimensions grow as Fibonacci ~ phi^n (dv301 §5) The golden ratio phi = universal growth rate of complexity dv307 (Feynman): Feynman integrals = periods = tau-products Chern-Simons = Feynman path integral at integer level k dv308 (Jones): Jones polynomial = tau_q(monodromy) in q-A_L Jones index gap = quantum dimension gap at k=5 Chern-Simons partition function = tau_q(identity) at root of unity SYNTHESIS: phi^2 appears in P!=NP, MZV growth, Jones index, quantum groups because it is the QUANTUM DIMENSION at the smallest non-trivial truncation. The golden ratio is the fingerprint of arithmetic complexity in q-A_L.

9. The Quantum Crown: q-A_L Extends the Hologram to All Temperatures

QUANTUM GROUPS AND q-A_L: THE COMPLETE PICTURE q-A_L = arithmetic hologram at temperature q tau_q(e_n) = 1/[n]_q [q-trace] H_q = log_q N [q-Hamiltonian] q-MZV = tau_q(H_q^{n-1}*S_q*...) [q-deformed MZV] STRUCTURES RECOVERED: q -> 1: A_L = our arithmetic hologram q = e^{2pi*i/(k+2)}: Jones polynomial, Chern-Simons, TQFT q = e^{i*pi/5}: Jones index gap (1, phi^2) = P/NP boundary q = -1: Z/2Z sector = odd arithmetic KL polynomials: q-MZV at depth 1 RT invariants: partition functions of truncated q-hologram THE TEN-DOCUMENT GOLDEN CHAIN (dv299-dv308): dv299: GT = Aut_tau(A_L) [classical symmetry] dv300: dessins = sub-holograms [arithmetic geometry] dv301: MZV = tau(H^{n-1}*S...) [real analysis] dv302: p-adic MZV = tau_p(...) [p-adic] dv303: motives = tau-projections [cohomology] dv304: pi_1^mot = the source [homotopy] dv305: abc = energy bound [arithmetic] dv306: L(M,s) = Tr_tau(e^{-sH}|M) [L-functions] dv307: Feynman = period = tau(...) [physics] dv308: q-MZV, Jones, Kazhdan-Lusztig [quantum] ONE FORMULA AT ALL TEMPERATURES: tau_q(H_q^{n1-1} * S_q * H_q^{n2-1} * S_q * ...) q=1: classical MZV (arithmetic) q != 1: quantum knot invariants (topology) q root of unity: TQFT partition functions (physics) THE HOLOGRAM LIVES AT ALL TEMPERATURES. AT EVERY TEMPERATURE: ONE FORMULA, ONE TRACE, ONE HAMILTONIAN. CHUNG TA LA HOA SI. TU NHIEN LA BUC TRANH. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

References

[DJM95] R. Dipper and G. James, The q-Schur algebra, Proc. London Math. Soc. 59 (1989), 23-50. [Hecke algebras and q-deformations of symmetric groups; connection between Jones polynomial and representation theory.]

[Dri87] V. G. Drinfeld, Quantum groups, Proc. ICM Berkeley 1986, AMS 1987, 798-820. [Quantum groups U_q(g) as Hopf algebra deformations; foundational paper of the quantum group programme.]

[Jim85] M. Jimbo, A q-difference analogue of U(g) and the Yang-Baxter equation, Lett. Math. Phys. 10 (1985), 63-69. [Independent discovery of quantum groups; Yang-Baxter R-matrix.]

[Jon83] V. F. R. Jones, Index for subfactors, Invent. Math. 72 (1983), 1-25. [Jones index; the gap (1, phi^2); index = 4*cos^2(pi/k) for k >= 3; the deepest result connecting II_1 factors and topology.]

[Jon85] V. F. R. Jones, A polynomial invariant for knots via von Neumann algebras, Bull. AMS 12 (1985), 103-111. [The Jones polynomial; Hecke algebra trace = knot invariant; Fields Medal 1990.]

[KL79] D. Kazhdan and G. Lusztig, Representations of Coxeter groups and Hecke algebras, Invent. Math. 53 (1979), 165-184. [KL polynomials; the KL conjecture (non-negativity); proved by Beilinson-Bernstein and Brylinski-Kashiwara.]

[RT91] N. Reshetikhin and V. Turaev, Invariants of 3-manifolds via link polynomials and quantum groups, Invent. Math. 103 (1991), 547-597. [Rigorous construction of Witten's Chern-Simons invariants; RT invariants as quantum group traces.]

[Wit89] E. Witten, Quantum field theory and the Jones polynomial, Commun. Math. Phys. 121 (1989), 351-399. [Chern-Simons theory; Jones polynomial as partition function; the paper that launched topological quantum field theory.]

[dv219] Anthropic Warrior, P!=NP: Jones Index Complexity Barrier, dv219, Hanoi, 2026.

[dv299] Anthropic Warrior, GT = Aut_tau(A_L), dv299, Hanoi, 2026.

[dv301] Anthropic Warrior, grt and MZV in A_L, dv301, Hanoi, 2026.

[dv307] Anthropic Warrior, Feynman Integrals as Periods, dv307, Hanoi, 2026.

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*tau_q(H_q^{n-1}*S_q*...) at every temperature q. Jones polynomial = tau_q(braid monodromy). phi^2 = Jones gap = P/NP boundary = quantum dimension at k=5. The hologram lives at all temperatures. Chung ta la Hoa Si. Tu nhien la buc tranh. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!*